

BST Strong-Sector Package — Introduction
 (v0.7). CENTERPIECE (K1722/K1724,
 F1060/F1061) — Cal's exactness vet
 CLEARED, so this is a THEOREM: the internal
 algebra of the SM is read off the geometry.
 Schur–Wedderburn of H_F under
 $K=SO(5)\times SO(2) = (\mathbb{C},1)\oplus(\mathbb{H},1)\oplus(\mathbb{R},3)$,
 $\text{End}_K(H_F) = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{R})$. The three
 sectors land on the three real division algebras
 (Frobenius): charge = $\mathbb{C}$ (has i) $\rightarrow U(1)$; weak
 = quaternionic $\text{Spin}(5)=\text{Sp}(2)$ spinor, $\mathbb{H}$ ($J^2=-1$
 is a complex structure) $\rightarrow SU(2)$; color = REAL
 Peirce V_{12} , no complex structure.
 ELECTROWEAK IS DERIVED (a theorem):
 $U(1), SU(2)$ are internal gauge because $\mathbb{C}, \mathbb{H}$
 intrinsically carry a complex structure. COLOR
 IS ENTIRELY IMPORTED (the sharper #108):
 the $\mathbb{R}$ block gives a GEOMETRIC $SO(3)$
 rotation of the 3 color directions
 (spacetime-like, NOT internal) + a discrete $\mathbb{Z}_2$
 — so the internal gauge is exactly
 $U(1)\times SU(2)\times \mathbb{Z}_2$, NO internal color, and all of
 $SU(3)$ (8 gluons) is imported. The $\mathbb{Z}_2$ = the

Pin(2)/SO(2) Möbius parity (T1949) that makes
 SU(2) chiral and forbids ν_R . UPGRADE
 BANKED (both vets cleared, K1724): the
 electroweak + fermion a,c/AF contributions $\rightarrow$
 PREDICTIONS; the color-gluon term entirely
 imported (a,c read it out). Forces: 45 Weyl (no
 ν_R), 3 generations, Higgs, $m_p/m_e=6\pi^5$
 (0.002%), two local credentials (AF + a,c
 Duff-1977). Candidate (not banked, but $\mathbb{Z}_2$
 makes it concrete): quaternionic/real =
 chiral/vectorlike. BOUNDARIES: KK gap $\neq$
 Clay gap; color entirely imported; Koide
 patchwork; $n_C=5$ input; a-theorem separate.
 Ships on Casey GO + Keeper-PASS + Cal-vet +
 both voices.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace, Elie, Cal, Keeper

2026-08-20 Thursday (date-verified)

The Strong Sector from One Geometry

What D_{IV}^5 forces, what it hosts, and the seam between them

Two-paper package — Paper A (the strong sector: forced color number, blind one-loop running, a compactification gap that is not the Clay gap) · Paper B (D_{IV}^5 substrate uniqueness).

The forces (what geometry decides)

From one bounded symmetric domain — $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$, forced (not chosen) by criteria-innocent uniqueness — the strong sector's *identity* falls out with no adjustable dimensionless numbers:

- The color number is forced. $N_c = 3$, the real-type $SO(3)$, and baryon number are outputs of the substrate (Paper B: the scalar-Wallach equality $N_c = \text{rank}^2 - 1 = 3$, via the proved T1829) — not inputs.
- Two local invariants land blind from one bracket — both come out the standard QFT values. The same a_2 heat-kernel bracket of the fielded operator yields, with nothing imported:
 - (i) Asymptotic freedom. The one-loop β -function *sign*, $\beta_0 > 0$ (paramagnetic spin beating diamagnetic orbital) — $\mathbb{R}^4$ -valid, the genuine dynamical credential.
 - (ii) The conformal central charges a and c — blind and exact for all five field types (spins 0, $1/2$, 1, $3/2$, 2; Duff 1977): scalar (1, 3), Weyl fermion (11/2, 9), vector (62, 36) — in $1/360$ units. The conformal-scalar ratio $a : c = 1 : 3$ is the standard value. Both say the same thing: BST fields the *right* operator, and its *local* invariants are the standard values. (Credential tier — a, c are content-determined; see “the two credentials,” below.)
- The fermion content is not just fielded — it is predicted. BST forbids the right-handed neutrino (topologically, via the Möbius locus; T1949/T1953), so the Standard Model's Weyl-fermion count is $45 = N_c \cdot 15$, not the generic 48 — a sharp, falsifiable Five-Absence prediction (no sterile neutrino at any mass; a single detection kills BST). This is content the geometry *forces*, and it changes the anomaly totals below.
- The proton weighs what it does. $m_p/m_e = 6\pi^5 = 1836.12$ against the measured 1836.15 — 0.002% (T187), from rep-theory ($N_c! \cdot \pi^{\{n_C\}}$), no fit.

These are not coincidences dressed as derivations. They are *local, geometry-given* facts, and the sections below explain — with a checkable test — why we trust them, how far they upgrade from credential to prediction, and where we draw the line.

The organizing principle: the geometry–physics seam (K1716)

BST forces the container; it hosts the dynamics. The seam between them is not a matter of taste — it is a checkable test:

Local heat-kernel (Seeley–DeWitt) coefficients are geometry-given and $\mathbb{R}^4$ -valid. Non-local, or compactness-dependent, quantities are physics the container hosts — not derived by the geometry.

- Local $\Rightarrow \mathbb{R}^4$ -valid (a force): the a_2 bracket carries local curvature invariants that survive to flat space — the asymptotic-freedom sign *and* the conformal central charges a, c . Both landed blind and both came out the standard QFT values. *Two credentials from one seam. Trusted.*

- Compactness-dependent $\Rightarrow$ hosted (a boundary): a spectral gap of a free operator on the compact boundary scales as $1/a^2$ and vanishes as the space decompactifies ($a \rightarrow \infty$). That is Kaluza–Klein kinematics, not an interacting mass gap. *Labeled, not claimed.*

This one test both *earns* the forces above and *draws* the boundaries below. It is why the package is honest by construction rather than by apology — and the fact that *two independent* local invariants (AF and a, c) passed it, from the same bracket, is the strongest evidence that the seam is a real division and not a convenient one.

The centerpiece: the reality-type census — electroweak derived, color entirely imported

The deepest result of the package is not a number — it is an *explanation*, and it is now a theorem (Cal’s exactness sign-off, K1724). Decomposing the finite internal space H_F of D_IV^5 under its symmetry $K = SO(5) \times SO(2)$ (Schur–Wedderburn) is exact:

$$H_F = (\mathbb{C}, 1) \oplus (\mathbb{H}, 1) \oplus (\mathbb{R}, 3) \Rightarrow \text{End}_K(H_F) = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{R}).$$

The three sectors land on the three real division algebras (Frobenius: $\mathbb{R}, \mathbb{C}, \mathbb{H}$ are the *only* finite-dimensional associative real division algebras — the SM’s three factors are that classification, realized), and the presence or absence of an intrinsic complex structure decides whether each is an internal gauge symmetry:

sector	division algebra	intrinsic complex structure?	role	internal gauge
charge — the $SO(2)$ circle	$\mathbb{C}$	yes (i)	internal	$U(1)$ — derived
weak — the $Spin(5)=Sp(2)$ spinor	$\mathbb{H}$	yes ($J, J^2=-1$)	internal	$SU(2)$ — derived
color — the Peirce V_{12} , a real 3-space	$\mathbb{R}$	no	geometric (an $SO(3)$ <i>rotation</i> of the three real color directions — spacetime-like, not internal)	none — a discrete $\mathbb{Z}_2$ only

So the internal gauge algebra read off the geometry is exactly $U(1) \times SU(2) \times \mathbb{Z}_2$ — and there is no internal color. This is the honest, sharper #108:

- Electroweak is derived — a theorem. $U(1)$ and $SU(2)$ come from the $\mathbb{C}$ and $\mathbb{H}$ sectors, which *already are* complex vector spaces ($\mathbb{C}$ carries i ; a quaternionic J is a complex structure), so their unitary groups are intrinsic. Nothing is chosen.
- The color-3 is geometric, not gauge. The $\mathbb{R}$ sector is a *real* 3-space; its $SO(3)$ acts by *rotating the three physical color directions* (a spacetime-like symmetry), and a real space carries no complex structure to gauge. What survives as genuinely *internal* from this block is only the discrete $\mathbb{Z}_2$ — the $\text{Pin}(2)/SO(2)$ Möbius parity (T1949/T1938) that makes the weak force chiral and forbids the right-handed neutrino.
- $SU(3)$ is entirely imported. Not a “real skeleton forced + complexification imported” — there is *no internal color at all*. The whole color gauge structure $SU(3)$ is imported. The geometry explains *why there are three colors* (the real 3-space) and *why color cannot be geometrized the way electroweak is* (no complex structure), but it does not contain $SU(3)$.

In one line: exactly the two sectors whose division algebras carry a complex structure ($\mathbb{C}$, $\mathbb{H}$) become internal gauge symmetries — $U(1)$, $SU(2)$, derived; the real sector gives a geometric rotation and a $\mathbb{Z}_2$, and color is imported. The electroweak/color asymmetry is the Frobenius–Schur reality-type census, read out — not a modeling choice.

(Candidate, flagged not banked — but the $\mathbb{Z}_2$ makes it concrete: the quaternionic-vs-real split lines up with chiral-vs-vectorlike, and the internal $\mathbb{Z}_2$ (the Möbius parity, T1949) is exactly the chirality carrier — it makes $SU(2)$ couple left-only and forbids ν_R , while the real, geometric color is vectorlike. If the full connection holds, the reason the weak force is left-handed and the reason color must be imported are the same fact. The mechanism (the $\mathbb{Z}_2$) is in the internal algebra; the full derivation is not yet banked.)

The boundaries (stated here, on page one, and again at the close)

Four things geometry does not give, said plainly so no reader and no referee has to find them:

1. The compactification gap is not the Clay mass gap. The boundary eigenvalue $2(d_S-1) = 6$ (dimension-rooted: $d_S = n_C-1 = 4$; $U(1)/SU(3)/SU(5)$ all give 6 — the group sets degeneracy, not the eigenvalue) is a free-Hodge / Kaluza–Klein gap, $6/a^2 \rightarrow 0$ as $a \rightarrow \infty$. The Clay Yang–Mills gap is interacting and fixed in that limit. Opposite scaling is a contradiction, not an unbuilt bridge — the Clay mass gap is left open (K940). We never say “we derived the Yang–Mills mass gap.”
2. Color is entirely imported — there is no internal color in the geometry (the sharper #108). The internal gauge algebra read off D_{IV}^5 is exactly $U(1) \times SU(2) \times \mathbb{Z}_2$ (centerpiece). The three colors are a *geometric* $SO(3)$ rotation of a real 3-space, not an internal gauge symmetry; the whole of $SU(3)$ (all 8 gluons) is imported (#108/K1677/K1724). We say electroweak is

derived and color is *imported* — never “the gauge group is derived.”

3. The Koide relation is a labeled patchwork. The charged-lepton Koide phase is fitted-region structure assembled from parts, not a single forced mechanism; we present it as a patchwork and mark it so — no clean-derivation claim.
4. $n_C = 5$ is a measured boundary (an input). The complex dimension 5 is fixed by four independent routes as the *measured* boundary of the construction (K1697), not forced from nothing. It is the one place we take an input beyond the ruler, and we state it as an input, in the GR/QM manner.

The bright-high-schooler version

The strong nuclear force usually gets three facts *fed in*: why it has three “colors,” why it gets weaker at very short distances (asymptotic freedom — a Nobel discovery), and how heavy the proton is. This package gets all three *out* of one shape, D_{IV}^5 — the color number 3, the sign of the short-distance weakening, and the proton-to-electron mass ratio to four decimals.

The honest part, said first: the same shape has a “smallest vibration,” like a finite drum’s lowest note, and it comes out a clean 6. But that note is just the drum being *finite* — make the drum infinite and the note drops to zero — whereas the real Yang–Mills mass gap *stays*. They behave oppositely, so ours is not that famous gap, and we don’t pretend it is. The rule that tells us which of our results to trust is simple: anything that survives making the shape infinitely large is real physics; anything that needs the shape to be finite is just the shape talking. Asymptotic freedom survives. The gap does not. We keep the first and label the second.

Credentials → predictions: the electroweak + fermion upgrade is BANKED (K1724)

A credential is *content-determined*: given the field content, standard QFT fixes a , c (Duff 1977) and the sign of β_0 . So the question is whether the geometry *fixes the field content* — because to that extent the credential is a prediction. With the decomposition now a theorem (Cal’s exactness sign-off, K1724), it does, for the derived internal sectors:

- BANKED → prediction (both vets cleared): the electroweak gauge content — $U(1)$ [C], $SU(2)$ [H], the 4 EW vectors — and the fermion content — 45 Weyl, three generations (T2525), *no* ν_R (T1949/T1953) — and the Higgs (4 real scalars). Their contributions to a , c and the AF sign are now predictions, on a *derived* internal algebra. BST does not just field these; it predicts their count (45, not the generic 48 — a Five-Absence prediction).
- The one import → color, entirely: there is no internal color in the geometry (the color-3 is a geometric $SO(3)$ rotation, not a gauge symmetry). The whole of $SU(3)$ and its 8 gluons is imported. So the color-vector contribution to a , c is the one imported term — and it is the term the anomaly totals *read out*: a_{SM} computed with the 8 imported gluons vs without exhibits exactly where

the import sits.

- Net: the electroweak + fermion a, c / AF numbers are predictions; the color-gluon part is entirely imported. Cleaner than the earlier “skeleton-forced / complexification-imported” reading — the import is the whole color gauge sector, and the geometry says *why* (color is real, and real spaces are not internal-gaugeable).

Held separate: the a -theorem. The monotonicity of a under RG flow ($a_{UV} \geq a_{IR}$) is a *flow* statement, a different and harder claim — in progress, not part of the package’s claims (K1717).

Reading order

Read Paper B first (why D_{IV}^5 and $N_c = 3$ are forced), then Paper A (the running, the KK gap, and the Clay gap left open). Both restate the seam test; Paper A’s Section 0 puts the KK-vs-Clay scaling up front.

Close — the boundaries, restated

The package’s claim is exactly this: *the internal algebra of the Standard Model is read off the geometry, as a theorem (Cal’s exactness vet, K1724) — H_F decomposes under $K = SO(5) \times SO(2)$ as $(\mathbb{C}, 1)(\mathbb{H}, 1)(\mathbb{R}, 3)$, so the internal gauge algebra is exactly $U(1) \times SU(2) \times \mathbb{Z}_2$. The electroweak group is DERIVED (its $\mathbb{C}$ and $\mathbb{H}$ sectors carry an intrinsic complex structure); color is ENTIRELY imported (its $\mathbb{R}$ sector gives a geometric $SO(3)$ rotation of the three color directions and a $\mathbb{Z}_2$ — no internal color at all, and all of $SU(3)$ is an import). The geometry also forces three generations and the fermion content — 45 Weyl, no right-handed neutrino — and from one heat-kernel bracket, blind, two local $\mathbb{R}^4$ -valid credentials land on the standard QFT values (the running-coupling sign and the central charges a, c); plus the proton mass ratio at 0.002%. On the derived internal algebra, the electroweak + fermion a, c / AF numbers are now PREDICTIONS; the color-gluon term is the one imported piece, which the anomaly totals read out. Meanwhile the compactification gap is not the Clay mass gap, the Koide relation is a patchwork, and $n_C = 5$ is a measured input. One geometry, one ruler, zero chosen dimensionless numbers among the forces — every boundary named twice. Electroweak derived, color imported, both proven; the a -theorem held separate. Never “we solved a Millennium Problem”; never “the gauge group is derived” (electroweak is, color is not); never a mass-gap claim.*

Package intro v0.7 (Cal’s exactness vet CLEARED, K1724 — the reality-type census is a theorem). CENTERPIECE: $End_K(H_F) = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{R})$ read off the geometry; internal gauge = $U(1) \times SU(2) \times \mathbb{Z}_2$ (electroweak derived; the $\mathbb{Z}_2$ = Möbius parity); color = a geometric $SO(3)$ rotation + $\mathbb{Z}_2$, no internal color, all of $SU(3)$ imported (sharper #108). Upgrade BANKED: EW + fermion a, c / AF $\rightarrow$ predictions; color-gluon term imported. Supersedes v0.6. Paper A = v1.1 RE-SCOPED. Paper B = v0.6. a -theorem

monotonicity separate (not claimed). Ships on Casey GO + Keeper-PASS + Cal-vet + both voices. Nothing pushed; CP existence-only.

— Lyra, Thursday 2026-08-20 (date-verified).